Predicting County-Level Diabetes Risk

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The Public Health Burden

- Type 2 Diabetes is a primary driver of cardiovascular disease and premature death.

Social Determinants of Health

- Community-level factors drive prevalence more than clinical markers alone.

Primary Project Objective

To identify and classify U.S. counties at the highest risk to guide intervention.

🎯 Target Audiences

- County public health officials
- Federal & state policymakers
- FQHC clinicians & community health workers

Defining the 'High-Risk' Class

≥ 10.4%

Threshold (75th %ile)

High-risk is defined as counties in the top 25% of diabetes prevalence, those exceeding 10.4% adult diabetes rates.

Data Integration & FIPS Mapping

📊 Data Integration

Merged CDC Diabetes Atlas + County Health Rankings into a unified dataset — 2,879 counties, 9 SDOH features.

🗺️ FIPS-Based Mapping

Federally assigned FIPS codes serve as the unique geographic key ensuring precise county-level data linkage.

Economic Predictors

- Median household income and childhood poverty are among the strongest risk markers identified by both models.

Education as Protection

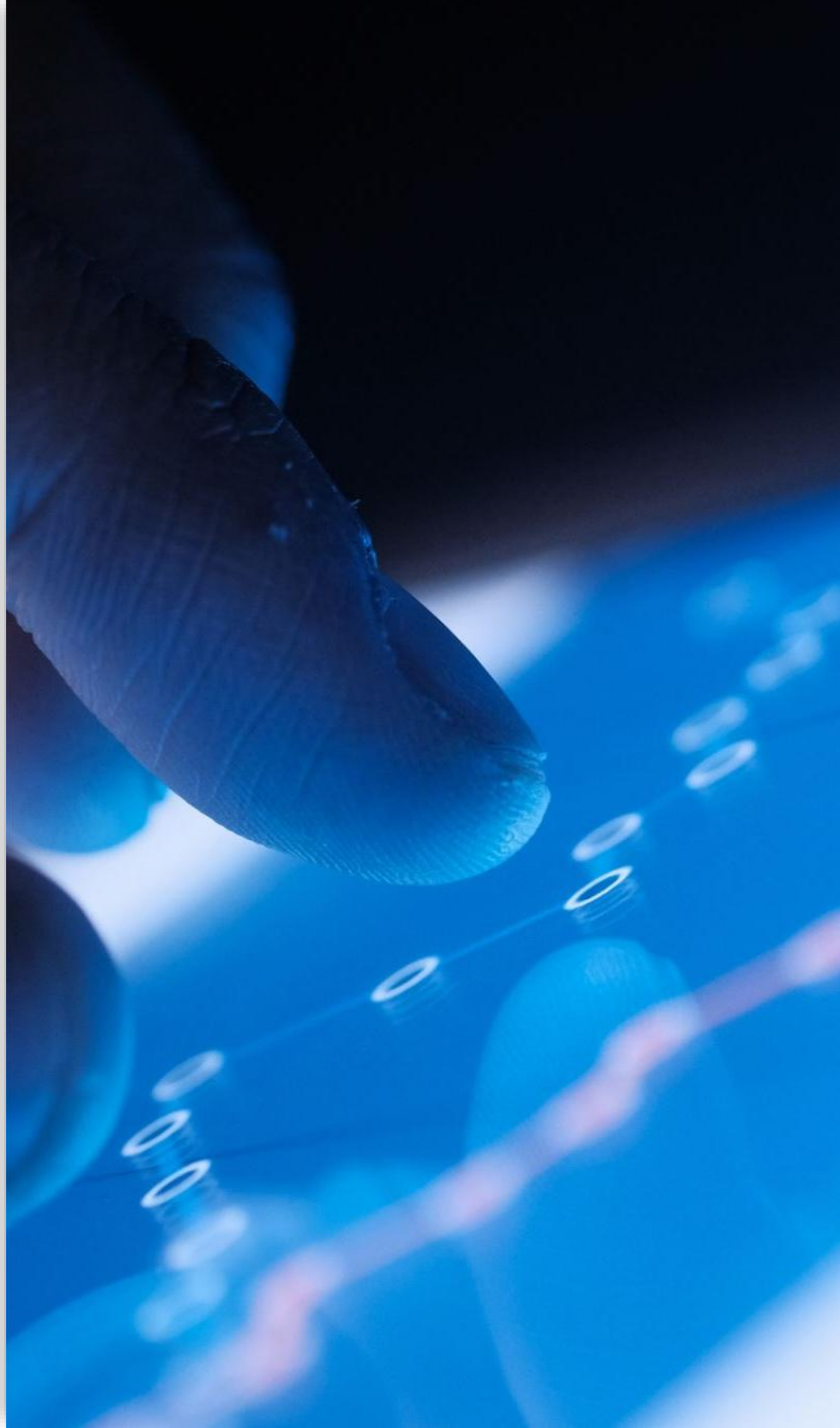
- Higher educational attainment (% some college) significantly lowers the odds of high diabetes prevalence.

Food Security and Housing

- Structural barriers like food deserts and housing cost burden are critical drivers of the disease burden.

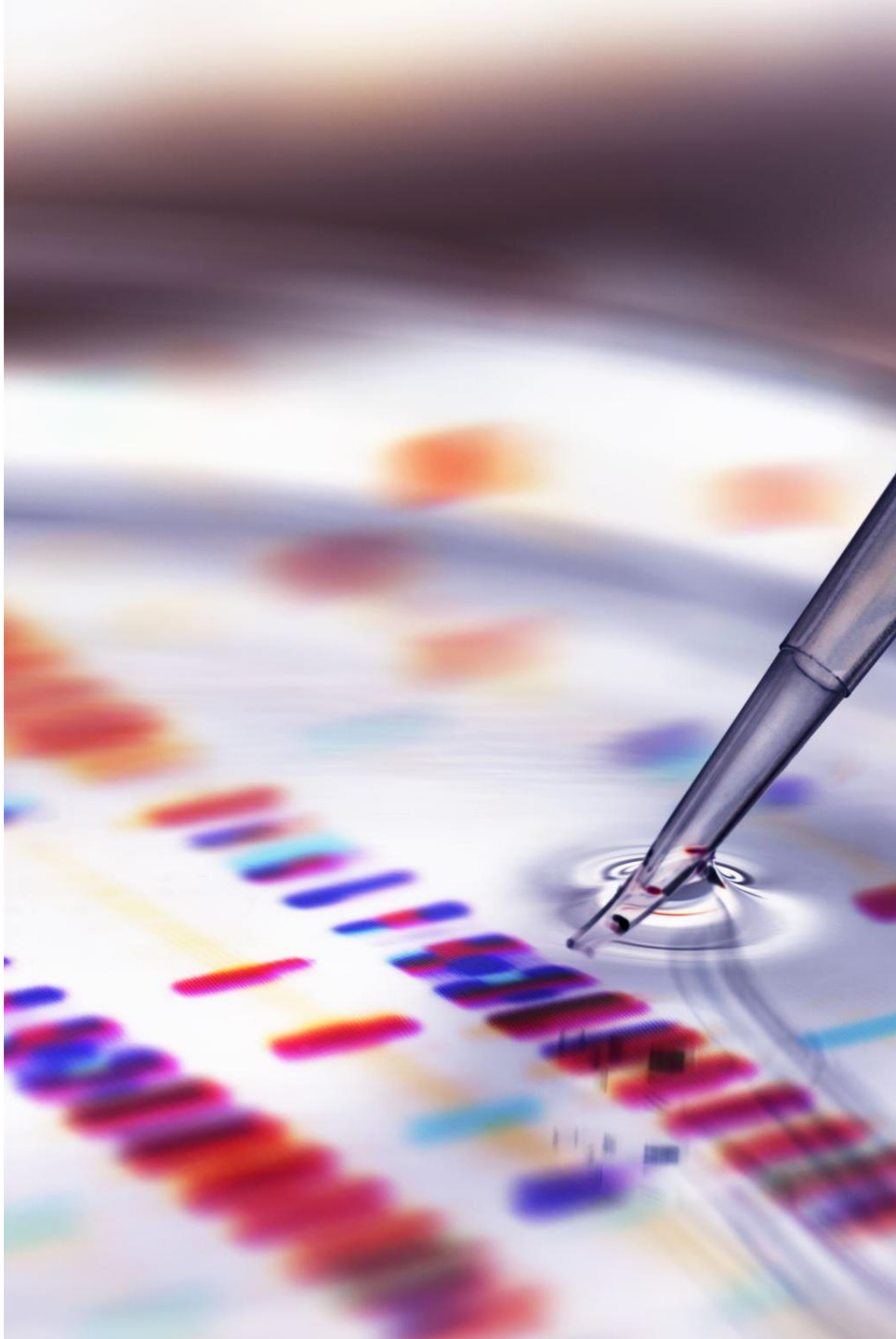
Data Preprocessing

- Data was standardized and scaled to ensure model accuracy and feature parity across all 9 SDOH variables.



Multicollinearity Analysis

- VIF analysis confirmed manageable multicollinearity among variables like income and education levels, all 9 features were retained.



LASSO Regularization

LASSO selected the 9 most impactful drivers by zeroing out irrelevant features, confirming all SDOH predictors were meaningful.

Logistic Regression Baseline

A baseline model provides interpretable odds ratios for health policy makers.

0.791	73.8 %	0.806

Model Performance: AUC-ROC Comparison

Logistic Regression

AUC-ROC	0.791
Accuracy	73.8%
Precision	50.0%
Recall	72.0%
CV AUC	0.806

Random Forest

★ Best Model

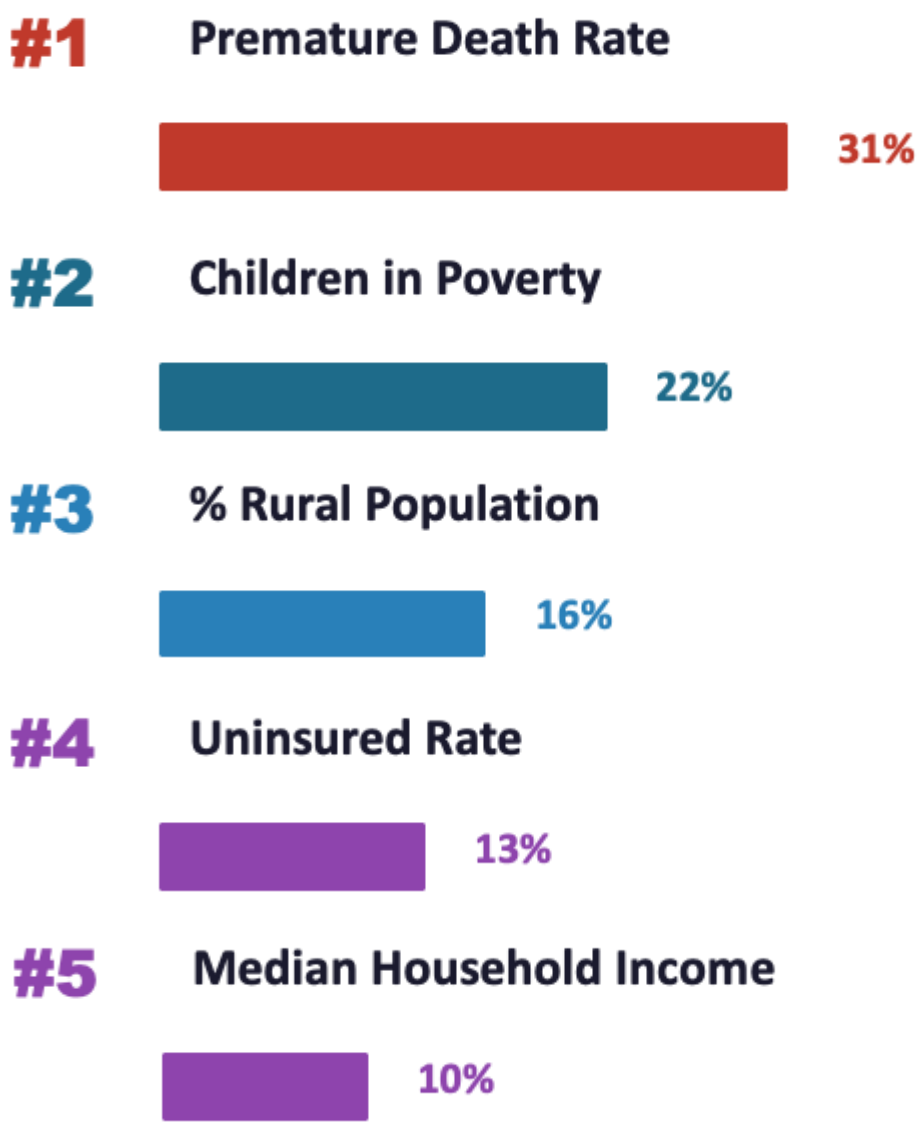
AUC-ROC	0.829
Accuracy	80.7%
Precision	66.0%
Recall	55.0%
CV AUC	0.827

Random Forest outperforms Logistic Regression across all key metrics.

Advanced Modeling: Random Forest

- Random Forest captures non-linear interactions between complex SDOH factors, achieving AUC = 0.829 and 80.7% accuracy.

Top SDOH Drivers of Diabetes Risk



Source: Random Forest feature importance, both models agree on top 3 predictors.

CDC HI-5 Framework

Our approach supports health impact within 5 years across all three HI-5 buckets:

Bucket 1
Healthy & Safe Environments

% Rural → Mobile clinics & telehealth

Housing Cost Burden → SNAP/housing vouchers

Bucket 2
Reduce Risky Behaviors

Children in Poverty → CDC National DPP

Unemployment → Workforce health programs

Some College → Community education outreach

Bucket 3
Clinical & Social Services

Premature Death → Medicaid expansion

Uninsured Rate → FQHC referrals

Median Income → Targeted subsidies

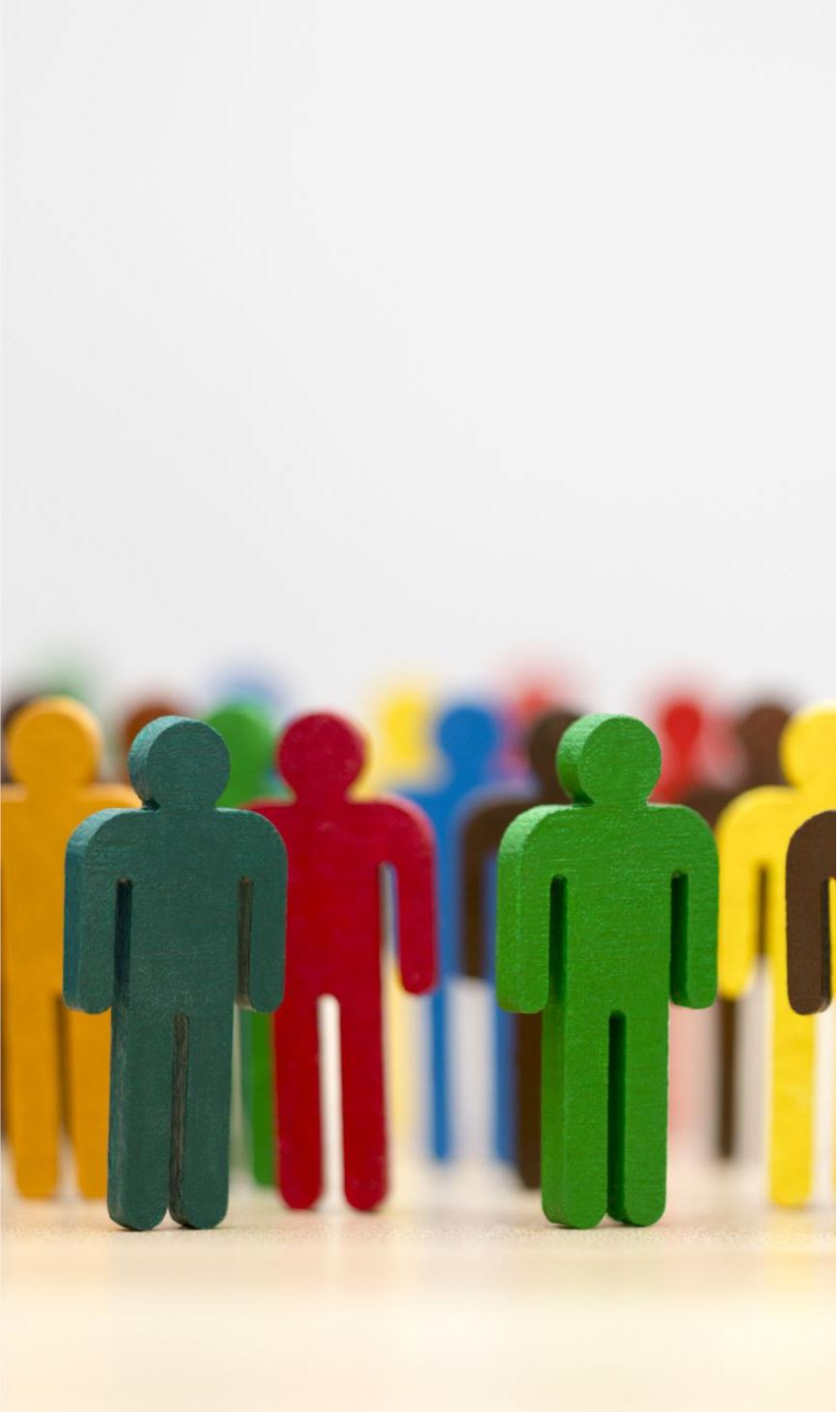
Surveillance Dashboard

A conceptual tool allows for annual monitoring of county risk clusters using updated FIPS data and model re-scoring.

🔄 Annual Data Pipeline (April)

① Ingest → ② Clean & FIPS Merge → ③ Feature Engineering → ④ RF Re-Score → ⑤ Dashboard + Alerts

Model retrained every 3 yrs or if AUC drops below 0.75



Policy Recommendations

📍 Deploy CDC National DPP in counties with high child poverty rates (top SDOH driver #2)

🏥 Expand Medicaid & FQHC outreach in counties with high premature death + uninsured rates (drivers #1 & #4)

🚐 Fund mobile health clinics and telehealth in high-rurality counties (driver #3)

📊 Prioritize the 753 identified high-risk counties for SDOH-targeted funding allocation

Integrating machine learning with SDOH creates a pathway for equitable health.

Conclusion

- 753 high-risk counties identified across the U.S.
- Random Forest AUC = 0.829, outperforming LR (0.791)
- Premature death, child poverty & rurality are top drivers
- CDC HI-5 framework guides actionable interventions